

My Vision as an Engineer

COOP 2100: Professional Praxis

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My vision as an engineer is built on a simple belief: the parts nobody notices are the ones that matter most. When I designed and built a sim racing shifter from scratch — sourcing Hall-effect sensors, machining interchangeable cam profiles, tuning it until the throw felt exactly right — I learned that the difference between something that “works” and something that feels right often comes down to details most people will never consciously register. A driver doesn’t think about the cam profile mid-race. They just feel whether the shift is clean or not. That’s the standard I want to hold myself to as an engineer: build things so well that their quality shows up as an absence of friction, not a feature someone has to point out.

I see the same principle at a much larger scale in the infrastructure work I’m doing now. Monorail and BMU track systems, method statements, O&M manuals — none of it is glamorous, and almost none of it is visible to the people who will eventually rely on it. But that invisibility is exactly the point. A building maintenance unit that runs safely for twenty years isn’t a feature anyone celebrates; it’s just something that quietly does its job because someone got the tolerances, the documentation, and the coordination right the first time. My vision is to be the kind of engineer who treats that invisible reliability as the actual goal — not a byproduct of chasing something flashier.

For my stakeholders — the site teams who build from my drawings, the facilities managers who maintain systems for decades after I’ve moved on, the public who will never know my name but will ride an elevator or trust a track system I helped document — this vision serves the common good by refusing to treat “good enough” as good enough. Precision at the small scale (a shifter that feels right) and precision at the large scale (a system that stays safe long after handover) come from the same instinct: care applied where no one is watching.

If I had to give this vision an image, it would be the shifter mechanism itself — small, precise, easy to overlook, and completely responsible for whether the whole machine feels right. That's the kind of engineer I want to be: the quiet, well-machined part that makes everything else work.